

What is assignment?

Give recommendations to make the game better

Super Math Saga Game:

- F2P game
- Solving math question to go to the new level
- Revenue from buying hints on levels
- How to make the game better?



An **AB** test to test a new game experience.

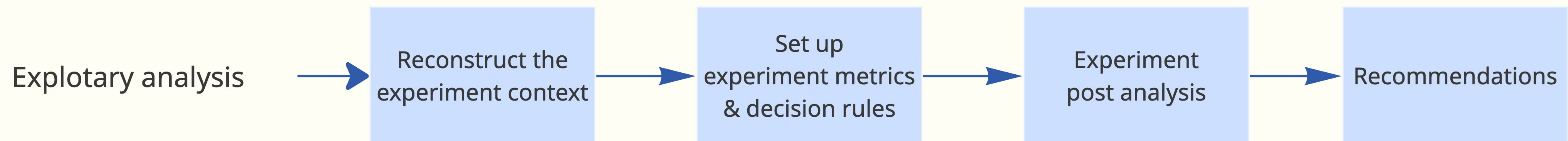
- Control group - A: same experience
- Treatment group - B: new experience
- Randomization at player level
- 2017-05-04 to 2017-05-22 (~2.5 weeks)

What is the challenging?

Missing the experiment context

- We need to understand the context (**goal, hypothesis, treatment**) to be able to find relevant insights & so give recommendations

How to approach?

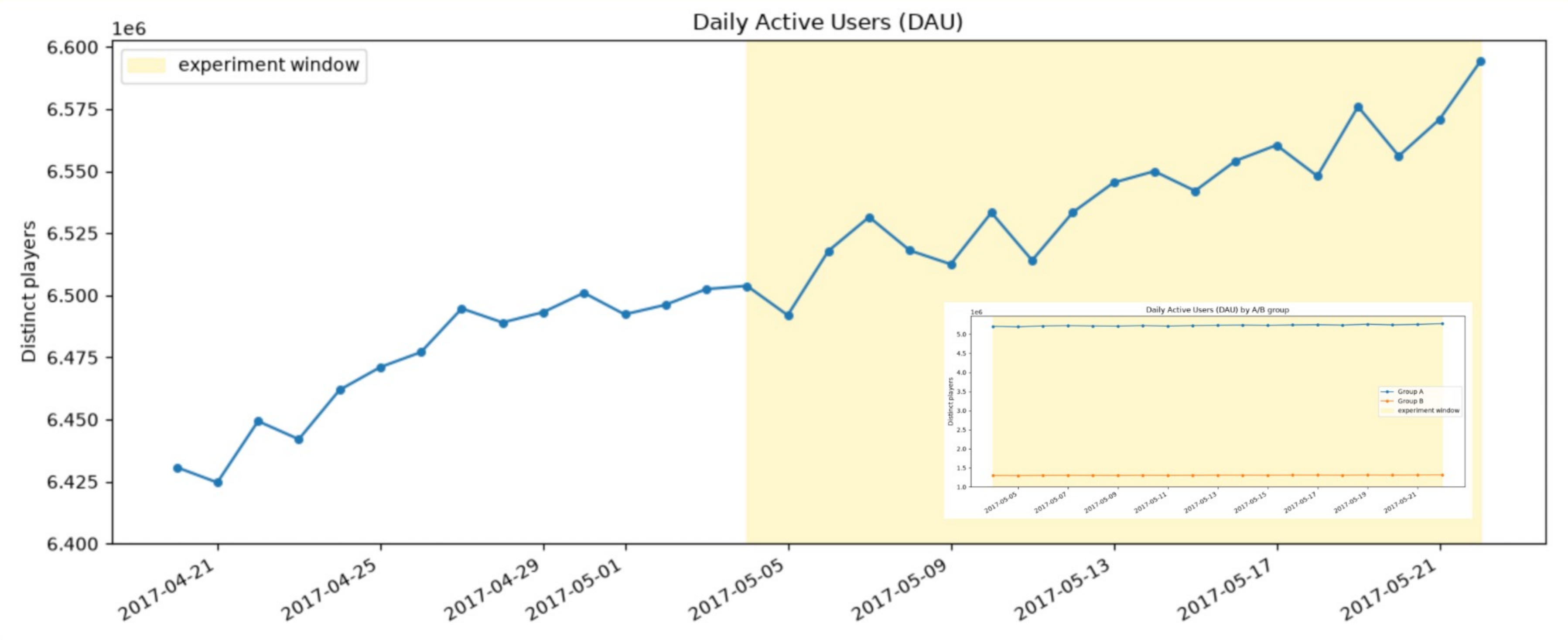


Available data: player ID, test group, active date, assigned date, first purchase date, game installed date, number of game rounds, number of purchase

The Experiment Context

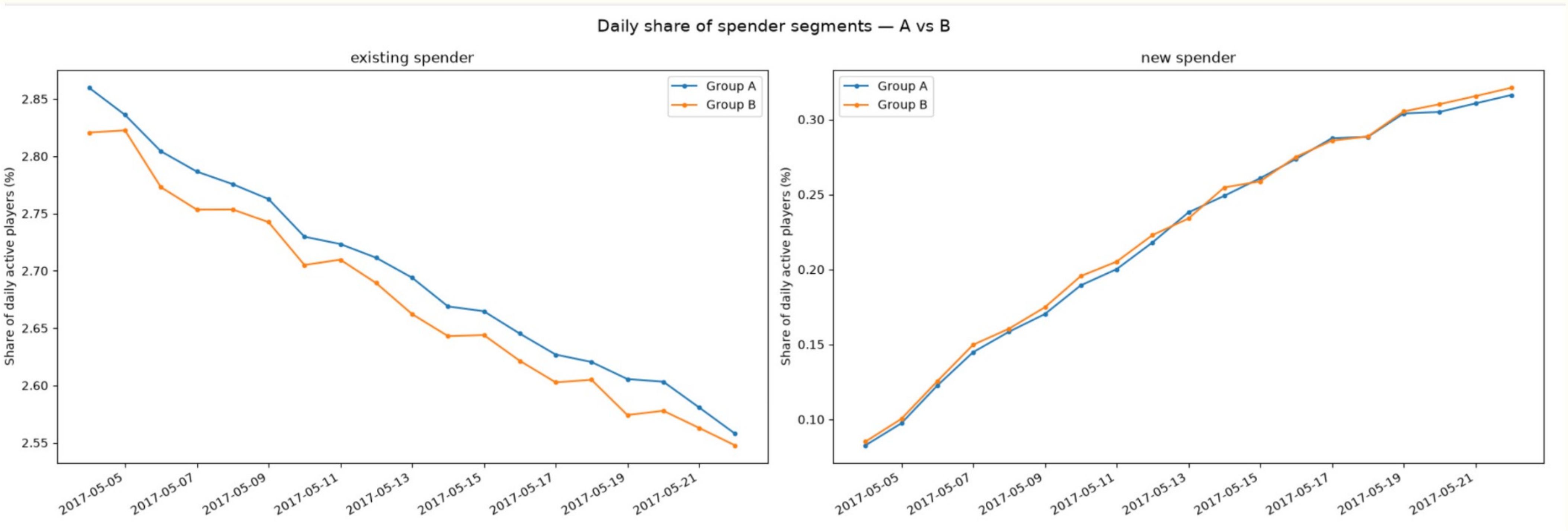
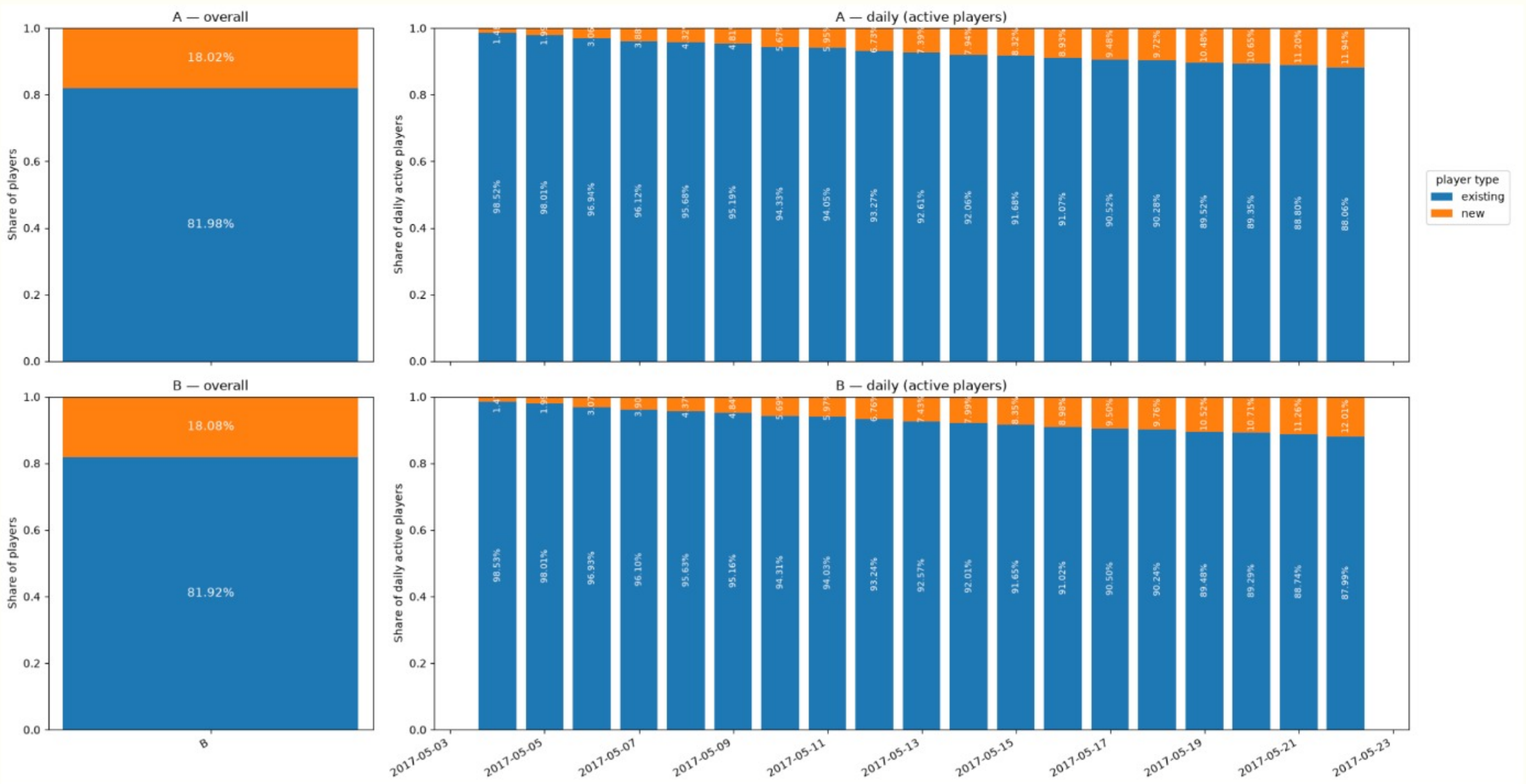
The test targets the general population. In practice, it was dominated by existing non-paying players.

- 1) 10.3M players during the test. DAU ~6.5 M
 - nearly 2/3 of users play every day. Stickiness seem too high to be realistic?
- 2) DAU kept increasing before the test and even during the ab test (+2.6%).
 - Not something the test caused.
- 3) AB test design ratio is kept closely at 80/20, seem no sign of SRM.
 - But need to a Chi-square test later to conclude



4) Exposure is broad and balanced across all user segments between Group A and B

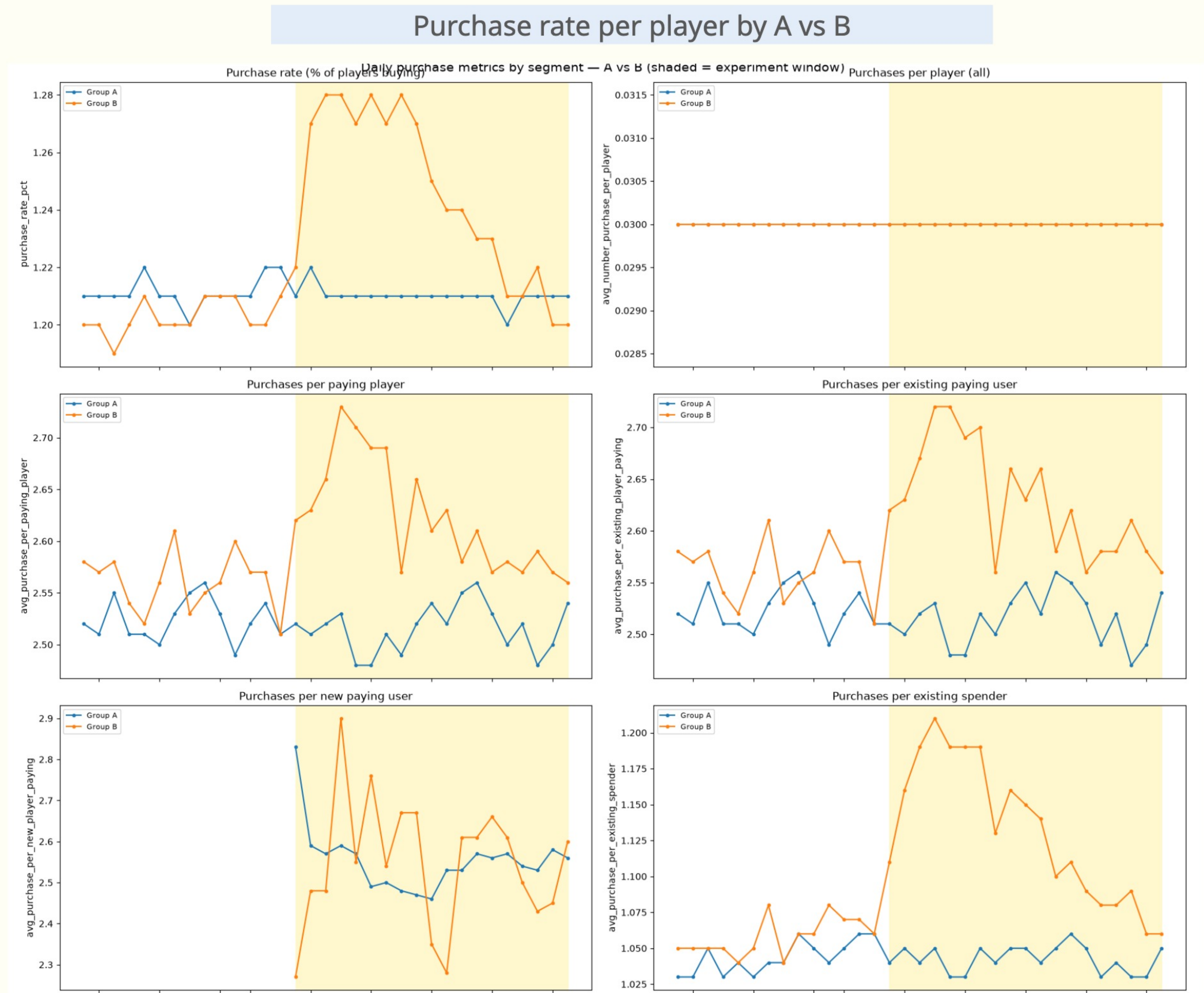
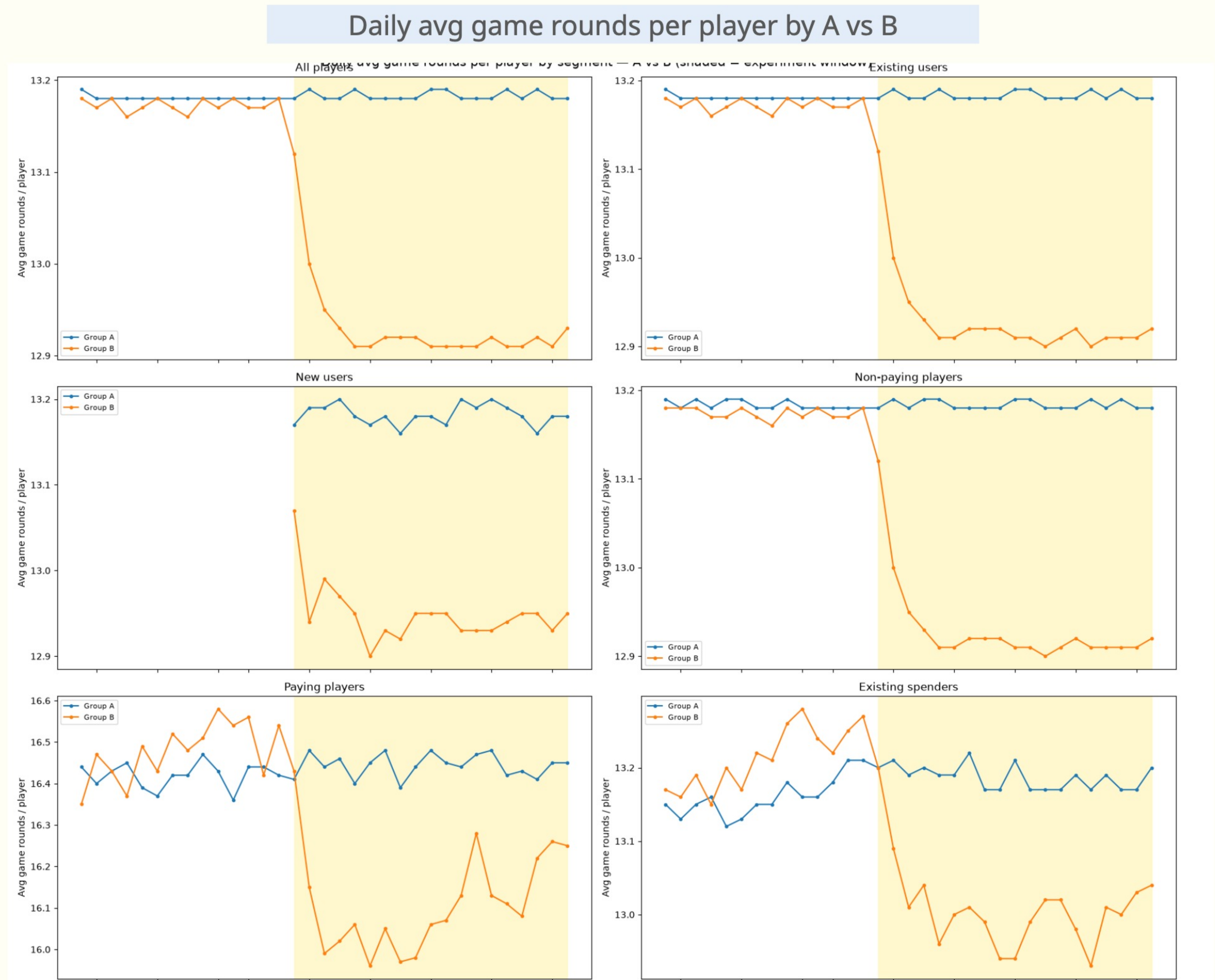
- new vs existing players
- spending players



The Experiment Context

From data, it shows that the test aims at increasing game revenue without hurting players' engagement.

- 1. After May 4, the treatment plays ~0.25 fewer game rounds/player (~1.9%) than control, across all segments. A is flat and the group were identical pre-test, so a broad engagement effect from the treatment.
 - A change to the game's difficulty/level mechanic ?
- 2. Before the test, the purchase rate of players in both group A and B are quite identical. After the test was started, the purchase rate for Group B was jumping from 1.21% to 1.28% then decreasing back (Novelty effect).
 - This indicates the game is harder in the treatment group that players, especially the existing spenders, buy more hints to pass the level. This makes their number of rounds decreased.



The Experiment Setup

Ship the new experience only if the primary metric shows a statistically significant, sustained lift and no guardrail metrics are declined.

Hypothesis Because hint demand rises when players struggle to progress unaided, we believe the **new experience in game difficulty will increase purchases per player while not harming the engagement** players.

- H_0 (null): The new experience has no effect on purchases per player vs. control ($B - A = 0$).
- H_1 (alternative): The new experience increases purchases per player vs. control ($B - A > 0$), one-sided.

**The new playing experience is that is harder to pass a level. It can be less free hints or increasing difficulty of math questions.
From the data (decrease number of game rounds while increasing the purchase), we can not differentiate what exactly the treatment is.*

Metrics

1. Success metric: capped purchases per player* (higher in B)
2. Guardrail metrics:
 - (uncapped) number of purchase per player
 - Number of game rounds per player
 - 7-day retention
3. Secondary metrics:
 - Purchase rate
 - Number of purchases per spender

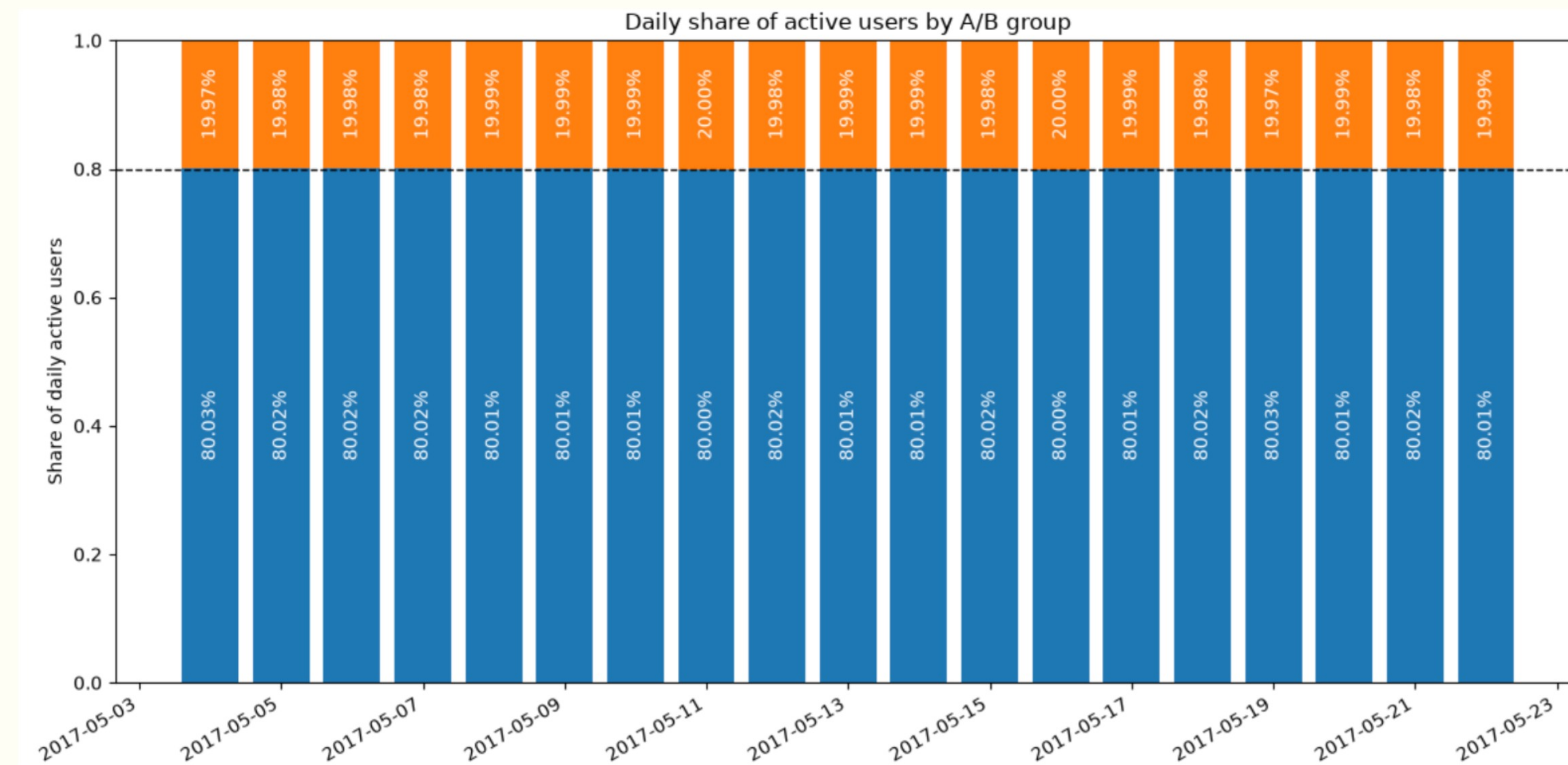
** Proxy metric due to data limitation (no revenue)*

p = 0.05 - CI 95%

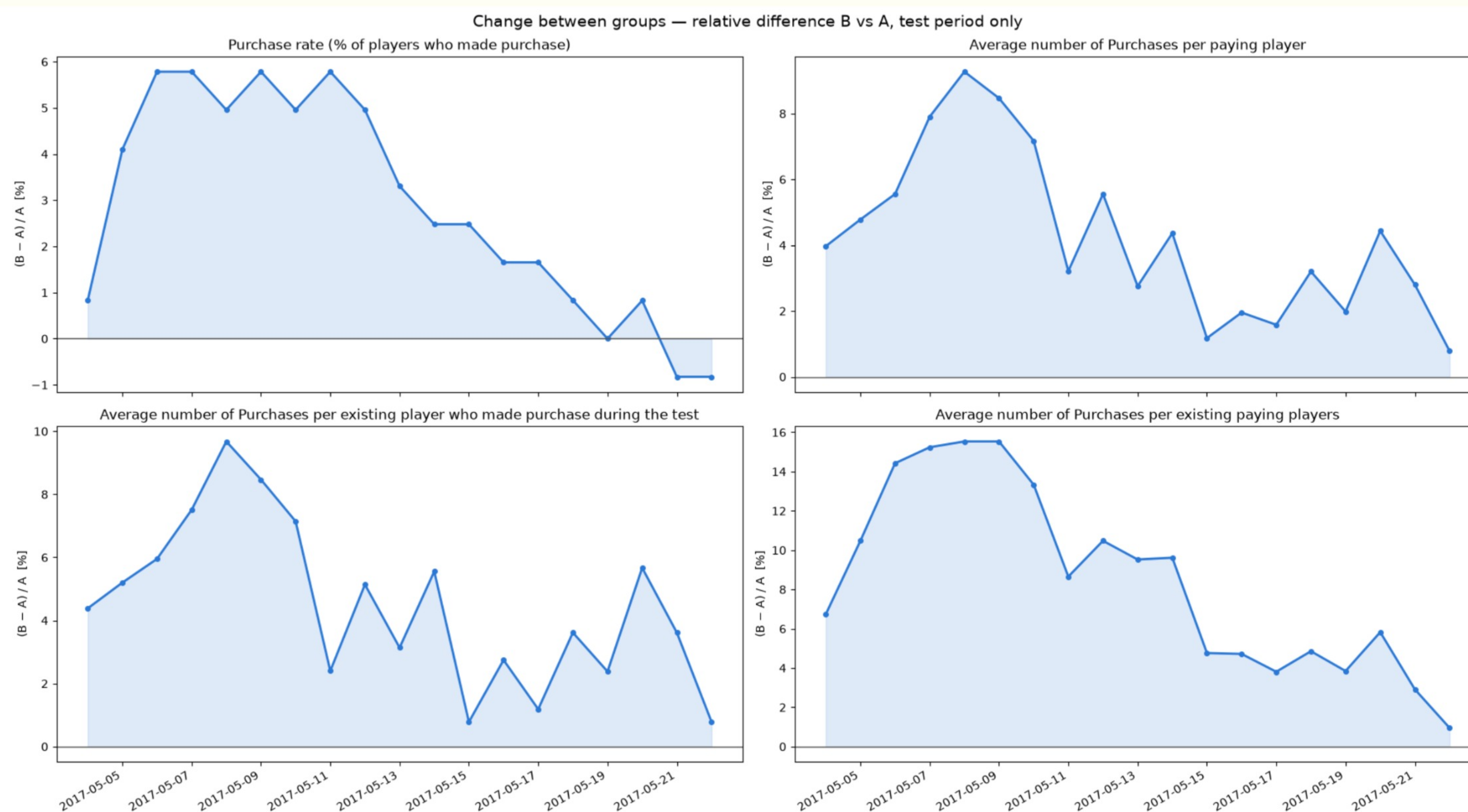
The Experiment Sanity check

No Sample Ratio Mismatch, but a clear observation of Novelty effect for the success metric!

- The test runned without any outbreak (DAU trend looks normal).
- One player belongs to one Group only.
- The users are randomized into Control and Treatment group according to the design ratio (**no SRM**, $p = 0.5517$, Chi-square test)
- Exposure is broad and **balanced across all user segments** (new vs existing users: non-paying vs spenders).



- **Novelty effect is observed** for the main related purchase metric. But the experiment was ended earlier that the difference in purchase between the two groups were not stabalized yet to observe a stable effect here.



The Experiment Results

Not trustworthy. So the result is inconclusive!



Success metrics

Capped number of purchases per users (p99)

- Control group: 0.298
- Treatment group: 0.31
- Uplift: +4.76% in the treatment
- Welch t-test:
 - CI95 [+0.00883, +0.01952]
 - $p \approx 1e-7$ ($\ll 0.05$)
- The direction is consistent across every segment, so it is not driven by one subgroup. However, **Novelty effect** was observed for the primary metric and the change was **not stabilized** yet when the test was ended. So the result for the **success metric is untrustworthy**.

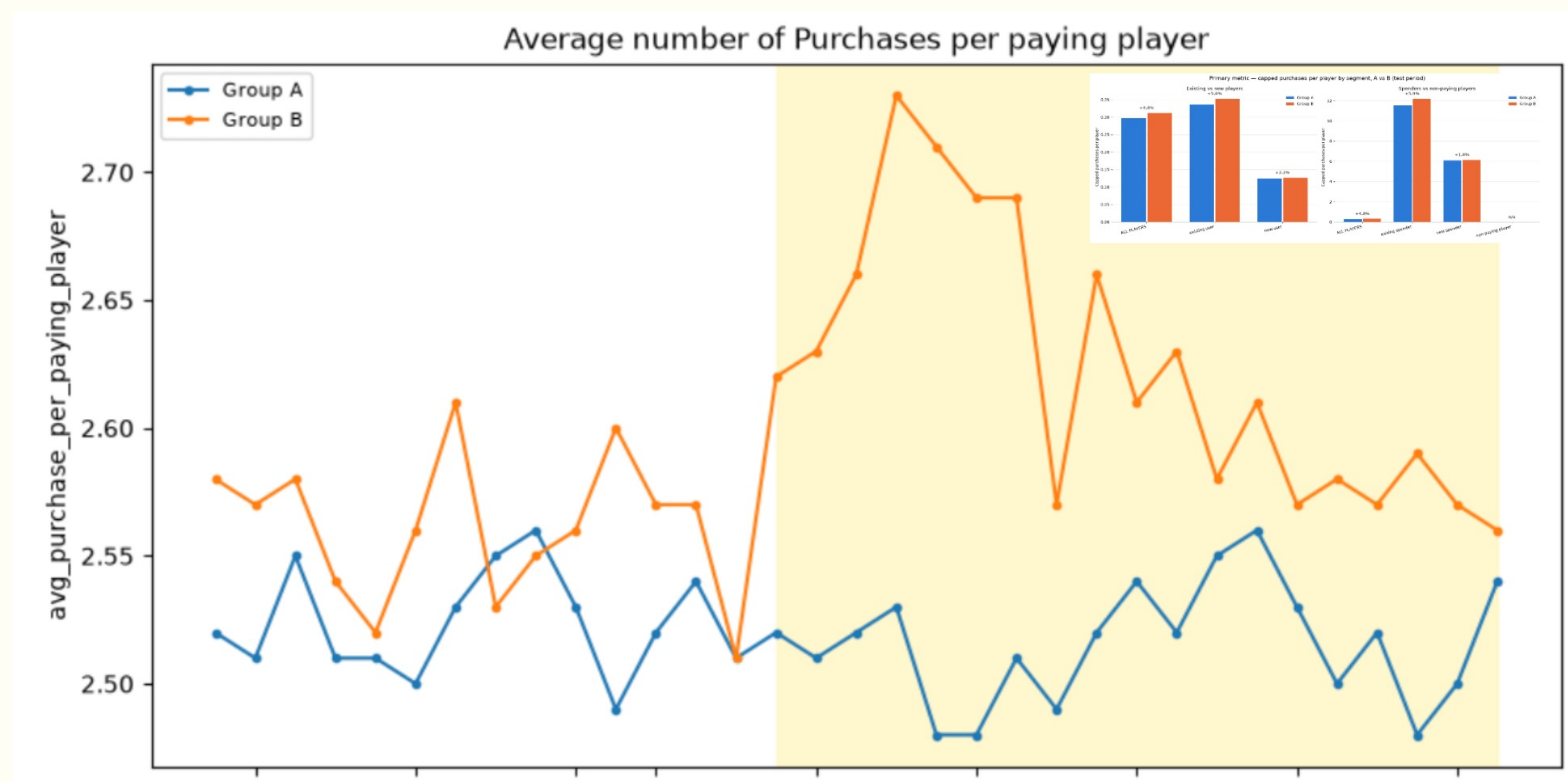
Guardrail metrics

Game rounds per player (engagement cost)

- Control group: 158.6
- Treatment group: 155.5
- Change: -1.94% (-3.07 rounds) in the treatment
- Welch t-test:
 - CI95 [-3.28, -2.86]
 - $p \approx 7e-182$ (one-sided, $H1: B < A$) -> **reject H0**.
- **B plays significantly fewer rounds**. Small in size but broad-based across every segment (-1.4% to -1.9%), and larger among non-payers (-1.9%) than spenders (-1.4%), i.e. genuine difficulty-driven disengagement, not just pay-to-skip.

7-day retention (active again within 7 days of assignment)

- Control group: 92.345%
- Treatment group: 92.320%
- Difference: -0.025 pp
- proportion z-test:
 - CI95 [-0.066, +0.015]
 - $p \approx 0.11$ (one-sided, $H1: B < A$) -> fail to reject H0.
- No sign of retention harm in-window.
- Note: frustration-driven churn is a slow, accumulating effect that this window cannot see.





Should we ship the new gaming experience?

Suggestion: Not ship the new gaming experience.

1. The experiment result (success metric) is not trustworthy. The result is inconclusive.
2. The guardrail metric for player engagement (number of gamerounds) is negatively effected.

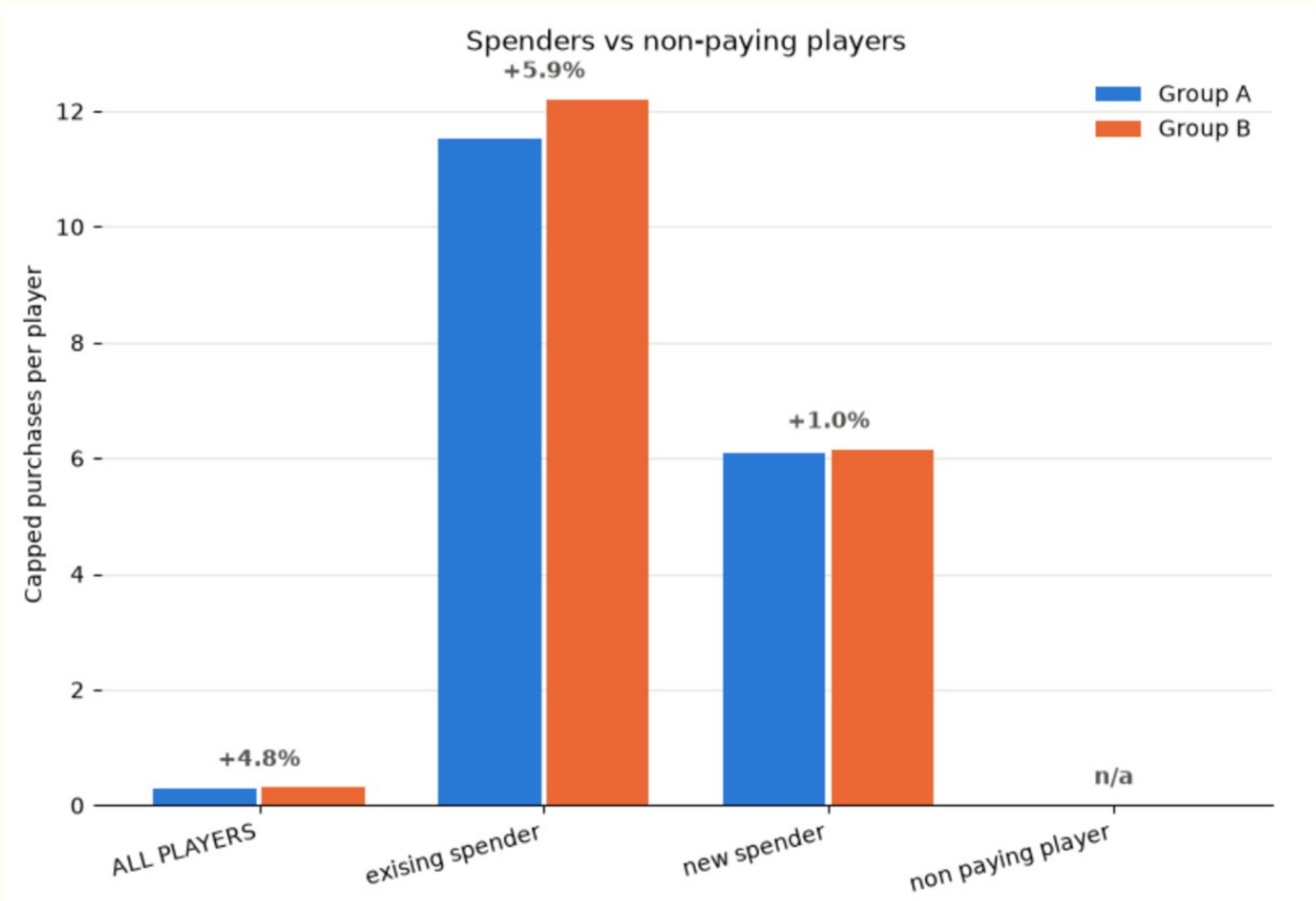
NOT SHIP the new gaming experience

Also not recommend to rerun the test longer to wait for the success metric leveled off. Because:

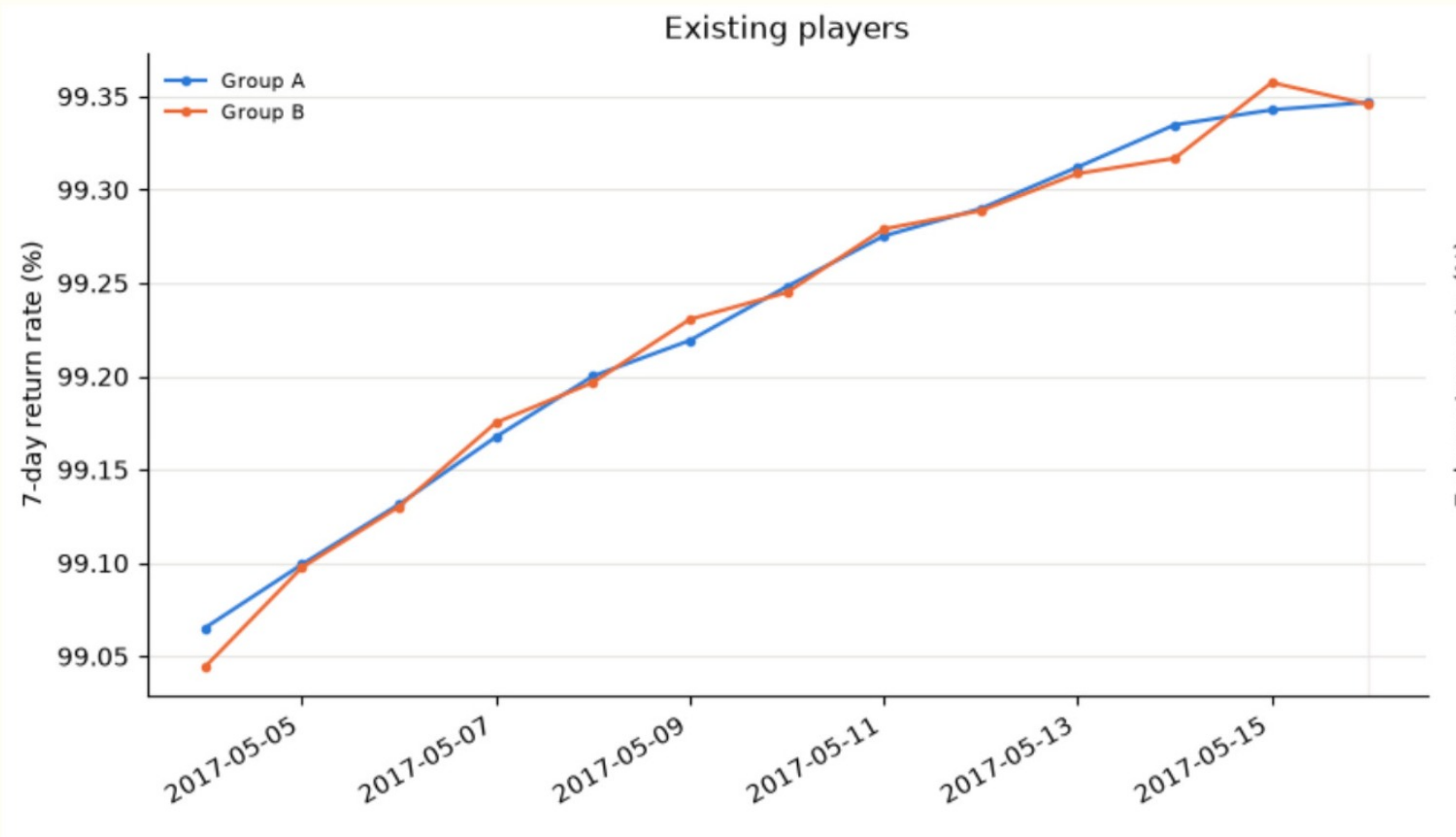
- the uplift effect was fading in the end of the test. So even with rerunning, it is suspected that the permanent change is low value
 - If non-paying users can not pass the test (decreasing number of gamerounds), they can become frustrated and can eventually quit. This shrinks the whole game community and can impact the attractiveness of the F2P game and so "paying players".
- => not good trade between "possible low /fading revenue uplift" for a permanent engagement cost.

What else can the product team do?

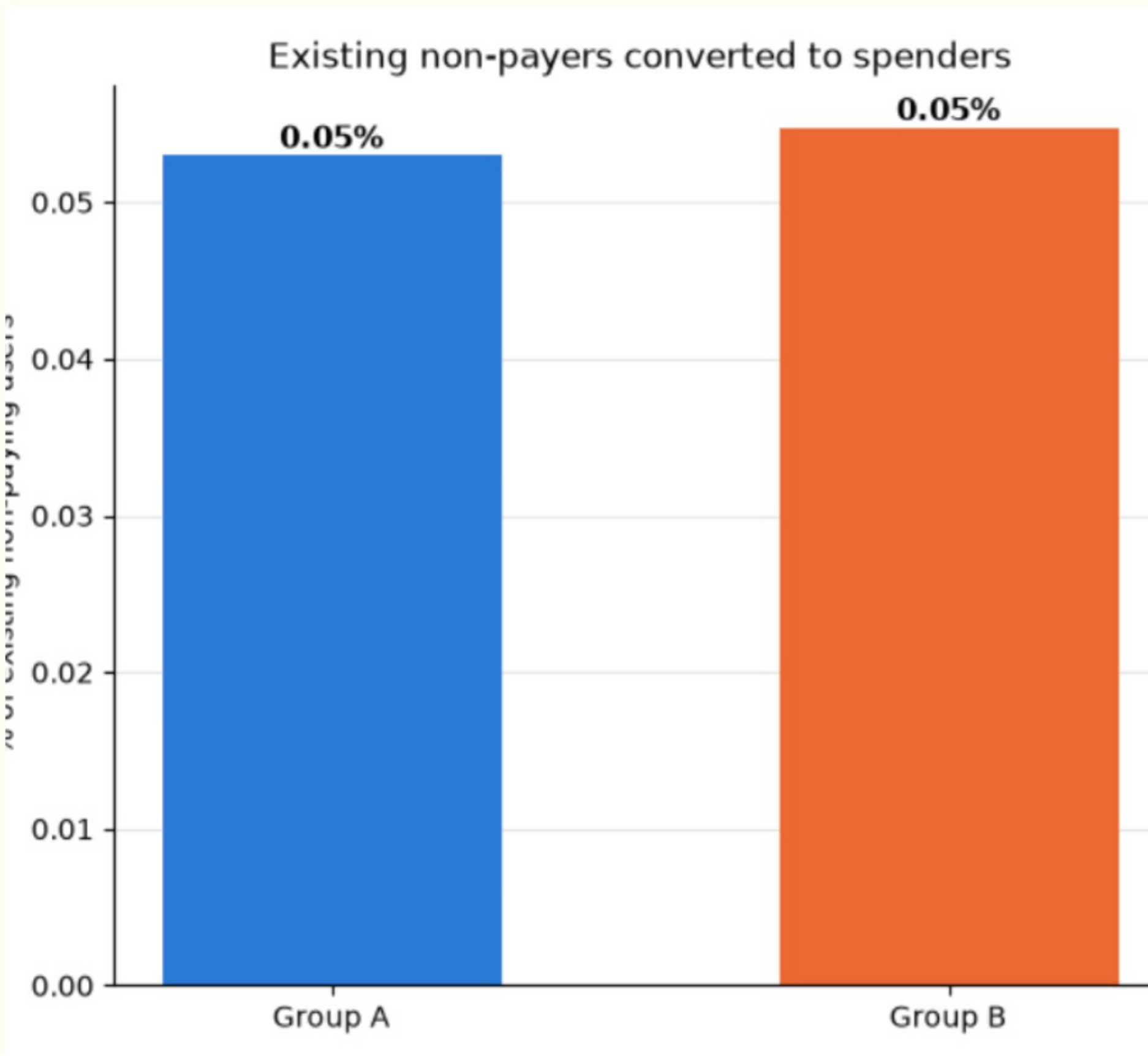
Monetize the Novelty effect: A math-solving challenging feature for players!



The uplift in purchases is intensity in existing spenders (+7%)



Retention (return to pay within a week) is same between Control and Treatment group



The more difficulty gaming experience did not grow the spender base

Hypothesis: difficulty can be deployed as a temporary/ time-limited event to boost revenue from existing spenders, without a permanent engagement cost.

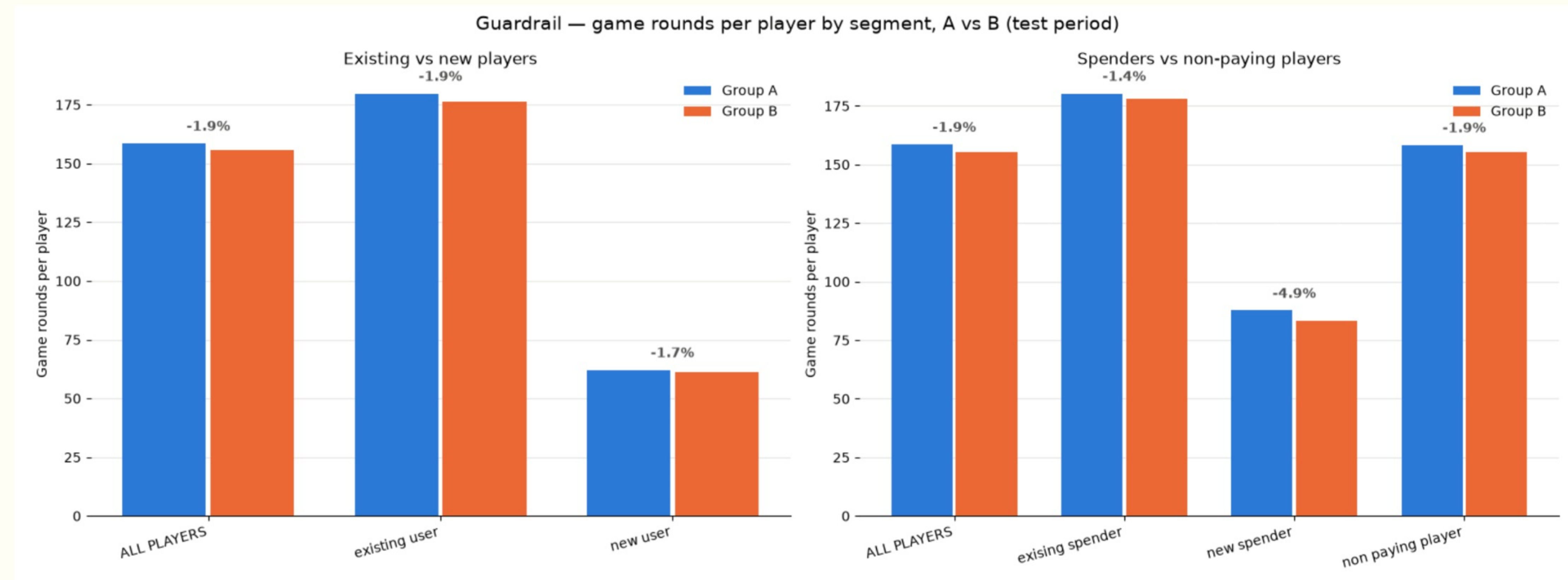
Propose: A math-solving challenging feature for players!



The Experiment Results: Appendix

Not trustworthy. So the result is inconclusive!

Guardrail metrics



- No Simpson paradox is observed.

Secondary metrics

Purchase rate (% of players who made a purchase):

- Control group: 2.690%
- Treatment group: 2.689%
- Difference: -0.03% (essentially zero)
- proportion z-test:
 - CI [-0.026%, +0.024%];
 - $p \approx 0.53$ -> **fail to reject H0**.
- The treatment did NOT get more players to buy: the size of the buyer base is unchanged.

Avg purchases per spender

- Control group: 13.61
- Treatment group: 14.60
- Uplift: +7.24% in the treatment group
- Welch t-test:
 - CI [+0.45, +1.52];
 - $p \approx 1e-4$ -> **reject H0**.
- Existing buyers bought more.